

Notes: DeMarzo (RFS, 2005)

The Pooling and Tranching of Securities: A Model of Informed Intermediation

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1 Big picture

- **Question.** Why do financial intermediaries pool assets and then carve the pool into tranches, rather than simply selling assets one by one or selling a pass-through pool?
- **Market friction.** The paper focuses on asymmetric information. The seller or intermediary may have superior valuation information about the assets, so buyers worry about adverse selection.
- **Key tension.** Pooling has two opposing effects. It destroys asset-specific information, but it can diversify residual risk and create a senior security that is less sensitive to private information.
- **Informed issuer result.** If an informed issuer can only sell pass-through claims, it prefers selling assets separately. Pure pooling is bad because it destroys the issuer's option to sell different quantities of high- and low-quality assets.
- **Tranching result.** If the issuer can create a derivative security backed by the pool, pooling and tranching can be optimal. A senior debt tranche can be made low-risk and information-insensitive when residual risks diversify.
- **Uninformed originator result.** If the seller is uninformed but some buyers are informed, pure pooling is useful. It reduces the ability of informed buyers to cherry-pick the best assets and therefore reduces underpricing.
- **Intermediation result.** These two forces generate a dynamic model of securitization: uninformed originators pool assets, informed intermediaries buy the best pools, and intermediaries then further pool and tranche to recycle capital.
- **Institutional motivation.** The paper is designed to explain asset-backed securities such as mortgage-backed securities, collateralized mortgage obligations, collateralized bond obligations, and other structured claims.
- **Economic takeaway.** Securitization is valuable not only because of transaction costs or market completion. It can be a way for informed intermediaries to turn private information into scalable capital turnover while selling low-information-sensitive claims to uninformed investors.

2 Data and measurement in the model

2.1 The asset payoff decomposition

The paper studies an issuer who owns n assets. Asset i has final payoff

$$Y_i = X_i + Z_i. \quad (1)$$

Here X_i is the issuer's private information about the expected value of the asset, while Z_i is residual risk.

- The issuer observes $X = (X_1, \dots, X_n)$.
- The residual risk satisfies

$$E[Z_i | X] = 0,$$

so $X_i = E[Y_i | X]$.

- The cumulative payoffs are

$$Y^n = \sum_{i=1}^n Y_i, \quad X^n = \sum_{i=1}^n X_i, \quad Z^n = \sum_{i=1}^n Z_i.$$

Interpretation. X_i is not accounting information; it is the seller's informational advantage. Z_i is the uncertainty that remains even after the seller's information is known.

2.2 Worst-case values and signaling

The paper assumes that each private-information component has a worst-case lower bound:

$$x_{i0} > 0. \quad (2)$$

For a pool, the corresponding worst-case value is

$$x_0^n = \sum_{i=1}^n x_{i0}. \quad (3)$$

Interpretation. The worst-case value pins down the separating equilibrium. A high-quality seller signals quality by retaining more of the asset, so buyers infer quality from how aggressively the seller sells.

2.3 The issuer's motive to sell

The informed issuer is risk neutral but discounts future asset cash flows at $\delta < 1$, while outside investors use the market discount rate normalized to one.

- The issuer prefers cash today because it has better uses for capital, such as buying newly identified underpriced assets.
- If the issuer sells fraction q of a pool at price p , its expected payoff is

$$E[\delta(1-q)Y^n + qp | X] = \delta X^n + q(p - \delta X^n). \quad (4)$$

Interpretation. The issuer wants to sell, but the act of selling is a signal. A high-quality issuer sells less because retaining the asset is less costly when the asset is valuable.

2.4 Why asset-backed securities are the natural application

- The model fits markets where specialized intermediaries have superior valuation ability.
- Examples include MBS, CMO, CBO, and other asset-backed securities.
- Even if many asset characteristics are public, valuation models and model outputs may be private information.
- The paper's motivating examples emphasize that intermediaries can value prepayment risk, credit risk, and pool quality better than many outside investors.

Interpretation. The model does not rely on hidden cash flows after issuance. The key asymmetry is ex ante valuation information held by the seller or intermediary.

3 Models and theoretical specifications

3.1 Informed issuer: selling a pass-through pool

Suppose investors have a demand schedule $P(q)$ for the pool. If $X^n = x$, the issuer chooses the quantity sold:

$$\max_q \delta x + q(P(q) - \delta x). \quad (5)$$

The profit from selling is

$$\Pi^P(x) = \max_{q \in [0,1]} q(P(q) - \delta x). \quad (1)$$

Interpretation. The seller's payoff is the discounted value of keeping the asset plus a trading surplus. The choice variable is how much of the claim to sell.

3.2 Convexity of the informed seller's payoff

For any demand function P , the payoff $\Pi^P(x)$ is decreasing and convex in x , and the optimal quantity sold is decreasing in x .

Interpretation. Convexity is the key economic force. Higher-quality sellers sell less; therefore the issuer has an option-like payoff from asset-specific information. Pooling averages away that option.

3.3 Separating equilibrium

In the separating equilibrium, the market price reveals the asset quality. For a worst-case value x_0 , the equilibrium takes the form

$$Q^*(x) = \left(\frac{x}{x_0}\right)^{1/(\delta-1)}, \quad P^*(q) = x_0 q^{\delta-1}. \quad (6)$$

The seller's equilibrium profit can be written as

$$\Pi^*(x) = \pi \left(\frac{x}{x_0}\right) x_0, \quad (7)$$

where $\pi(\cdot)$ is decreasing and convex.

Interpretation. Since $\delta < 1$, the exponent in $Q^*(x)$ is negative, so high-quality issuers sell smaller fractions. The market rationally interprets low sale quantity as good news.

3.4 Separate sales versus pure pooling

If assets are sold separately, the payoff is

$$\sum_{i=1}^n \pi \left(\frac{X_i}{x_{i0}}\right) x_{i0}. \quad (8)$$

If assets are pooled and sold as a pass-through, the payoff is

$$\pi \left(\frac{X^n}{x_0^n}\right) x_0^n. \quad (9)$$

Theorem 1 states that

$$\sum_{i=1}^n \pi \left(\frac{X_i}{x_{i0}}\right) x_{i0} \geq \pi \left(\frac{X^n}{x_0^n}\right) x_0^n, \quad (10)$$

with strict inequality unless all quality ratios X_i/x_{i0} are the same.

Interpretation. Pure pooling is worse for an informed issuer because it combines high- and low-quality assets into a medium-quality pool. This destroys the asset-level signaling option.

3.5 Information destruction effect

The cost of pure pooling is the **information destruction effect**. It follows from the convexity of the informed issuer's payoff in private information.

Interpretation. Pooling is not always good. It is harmful when the seller knows asset-specific quality and can exploit that information through selective sale quantities.

3.6 Tranching and security design

Now the issuer can create a security or tranche $F(Y)$ backed by the asset, where

$$0 \leq F(Y) \leq Y. \quad (11)$$

If the security's expected payoff conditional on X is

$$f = E[F(Y) | X], \quad (12)$$

and its worst-case value is f_0 , then the ex ante security-design problem is

$$G[Y] = \max_F E \left[\pi \left(\frac{f}{f_0} \right) f_0 \right]. \quad (2)$$

Interpretation. The issuer wants a security with a high worst-case payoff f_0 but low information sensitivity f/f_0 . This is exactly why senior debt can be useful.

3.7 Debt as the optimal monotone security

Under the paper's log-concavity conditions on residual risk, the optimal monotone security is standard debt:

$$F^*(Y) = \min(d, Y). \quad (13)$$

Interpretation. Debt pays the safest cash flows first. That makes its value less sensitive to the issuer's private information than equity or a pass-through claim.

3.8 Pooling and tranching

To analyze diversification, the paper uses a one-factor residual risk structure:

$$Z_i = \varepsilon_i + \eta, \quad (14)$$

where ε_i is idiosyncratic residual risk and η is common residual risk.

If the issuer pools the assets and issues a debt tranche, the payoff is

$$G \left[\sum_{i=1}^n Y_i \right]. \quad (15)$$

If it tranches assets separately, the payoff is

$$\sum_{i=1}^n G[Y_i]. \quad (16)$$

Interpretation. Pooling is useful only if residual risk diversifies. Tranching converts that diversification into a senior claim that is liquid despite private information.

3.9 Large diversified pools

Theorem 2 states that if residual risk is idiosyncratic and bounded, then as $n \rightarrow \infty$,

$$G\left[\frac{1}{n}\sum_{i=1}^n Y_i\right] \rightarrow (1-\delta)x_0, \quad D^*\left[\frac{1}{n}\sum_{i=1}^n Y_i\right] \rightarrow x_0. \quad (17)$$

Thus, pooling and tranching is optimal for sufficiently large n .

Interpretation. The senior debt tranche approaches a near-safe claim on the worst-case per-asset value. It becomes highly liquid because its value is almost independent of the issuer's private information.

3.10 No diversification, no benefit

Theorem 3 states that if residual risks are perfectly common, pooling is not optimal.

Interpretation. Pooling does not mechanically improve liquidity. It helps only when residual risks diversify enough to outweigh information destruction.

3.11 Information commonality

Theorem 4 states that the payoff from pooling and tranching increases when the private information in the pool is more common or general across assets.

Interpretation. Pooling is best when private information is about common factors and residual risk is asset-specific. This explains why pools are diversified geographically or by borrower, but not always across unrelated asset classes.

3.12 Multiple tranches and ex post design

With many tranches, selling a portfolio of tranches is equivalent to choosing a security design after observing $X = x$. The ex post problem is

$$\Gamma^P(x) = \max_F P[F] - \delta E[F(x + Z)]. \quad (3)$$

Under monotonicity conditions, the equilibrium design is again debt:

$$F_x(Y) = \min(d(x), Y), \quad (18)$$

with the face value $d(x)$ decreasing in x .

Interpretation. A lower-quality issuer sells more senior claims. Investors interpret a larger debt issue as a worse signal about the underlying pool.

3.13 Uninformed originators and adverse selection

An uninformed originator sells assets to a market with both informed and uninformed buyers. Uninformed buyers bid a price that breaks even, taking into account that informed buyers purchase high-quality assets.

The equilibrium uninformed bid is

$$P^*[X] = \max\{p : E[Q^U(X/p, p)(X - p)] \geq 0\}. \quad (4)$$

Theorem 6 states that if X_i are independent and the pool is large,

$$\frac{1}{n}P^*\left[\sum_{i=1}^n X_i\right] \rightarrow E[X]. \quad (19)$$

Interpretation. Pooling helps uninformed sellers because informed buyers cannot isolate only the best assets. The pool becomes less vulnerable to cherry-picking.

3.14 Dynamic intermediation

The intermediary starts period t with cash C_t and old securities worth O_t . It buys new underpriced assets with full-information value N_t and may have unused cash U_t . It then sells old securities and part of the new assets, generating sale proceeds S_t .

The next-period balance sheet is

$$C_{t+1} = U_t + S_t + O_t, \quad O_{t+1} = N_t - S_t. \quad (20)$$

Interpretation. The intermediary's core problem is capital turnover. If it can resell assets or senior tranches without suffering too much adverse selection, it can use the cash to buy more underpriced assets in the next period.

3.15 Acquisition with cash constraints

In the origination market, the informed intermediary buys assets above a cutoff x^c . If it has cash C , market size M , and participates in fraction θ of sales, the cutoff solves

$$x^c = \min x' \quad \text{such that} \quad x' > p \quad \text{and} \quad \theta M \Pr(X > x')p < C. \quad (5)$$

Uninformed investors break even at price p :

$$E[(1 - \theta \mathbf{1}\{X > x^c\})(X - p)] = 0. \quad (6)$$

Interpretation. More intermediary cash makes informed buying more aggressive. This worsens adverse selection for uninformed buyers and lowers the market bid.

3.16 Growth with securitization

Theorem 7 states that, given initial assets, the intermediary is larger at future date $T > 2$ when pooling and tranching is possible than when assets are sold individually or not resold, provided that the intermediary has a cash motive in some earlier period.

Interpretation. Securitization is not just a static pricing innovation. It changes the growth rate of informed intermediaries by allowing them to recycle capital.

4 Conceptual design and what makes the argument credible

4.1 The central identification is theoretical

This is a theory paper. Its logic is not based on estimating causal effects in data. Instead, it identifies the economic forces by comparing institutional arrangements in a formal model:

- separate sale;
- pure pooling;
- pooling plus tranching;
- uninformed originator sale;
- dynamic informed intermediation.

Interpretation. The paper's credibility comes from showing that the same asymmetric-information framework produces different predictions for informed and uninformed sellers.

4.2 Why pure pooling hurts informed sellers

- Informed sellers have asset-level information.
- They can signal high quality by selling less of a high-quality asset.
- The seller's value from this quantity choice is convex in asset quality.
- Pooling averages the qualities and removes the ability to condition sale quantity on each asset.

Interpretation. This is why the paper rejects the simple idea that pooling always reduces asymmetric-information costs.

4.3 Why tranching reverses the conclusion

- Tranching creates a senior security with low information sensitivity.
- Pooling diversifies residual risk, making the senior security safer.
- A large diversified pool can support a high-quality debt tranche.
- The issuer retains the information-sensitive residual.

Interpretation. Tranching allows the seller to sell what is liquid and retain what is information-sensitive.

4.4 Why uninformed sellers pool

- The uninformed seller cannot signal asset-specific quality.
- Informed buyers can otherwise select only the highest-quality assets.
- Pooling reduces the precision of informed selection.
- The uninformed bid approaches the expected value as the pool grows.

Interpretation. The same pooling decision has opposite implications depending on who holds the information.

4.5 What the model cannot fully prove

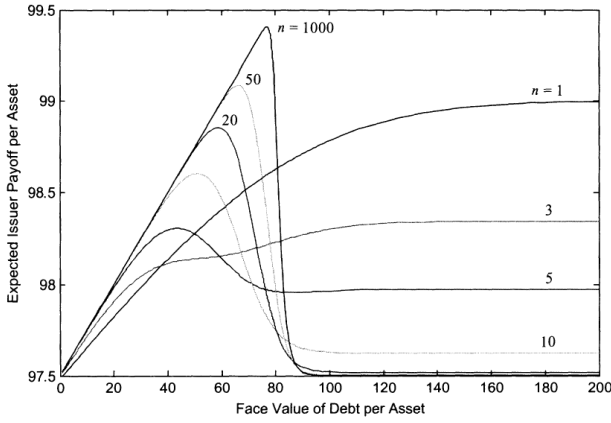
- The paper does not estimate the model on transaction-level securitization data.
- It abstracts from many real-world frictions, including taxes, regulatory capital rules, ratings, legal structuring costs, and risk aversion in the main model.
- It restricts attention to monotone securities for tractability.
- It assumes stylized forms of information and residual risk.
- It focuses on asymmetric information, while real securitization markets also involve transaction costs, incomplete markets, accounting treatment, and regulation.

Interpretation. The paper is best read as a clean mechanism paper: it isolates why pooling plus tranching can be valuable under information asymmetry.

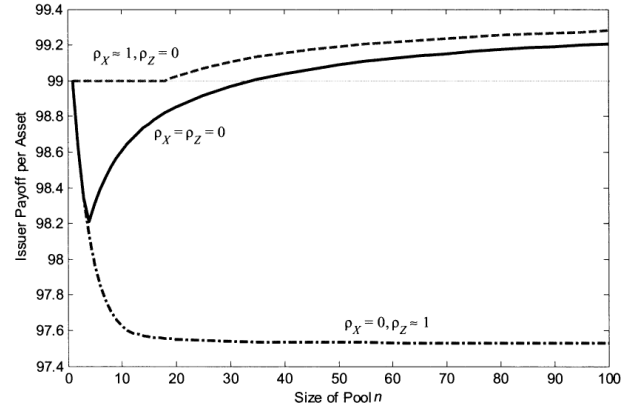
5 Figures

The paper has no empirical tables. Its key visual material is a sequence of theoretical graphs and diagrams. The figures below are grouped to save space and improve comparison; all images are directly cropped from the original paper.

5.1 Figure 1 + Figure 2: Pool size, debt face value, and correlation



(a) Figure 1: payoff by debt face value and pool size.



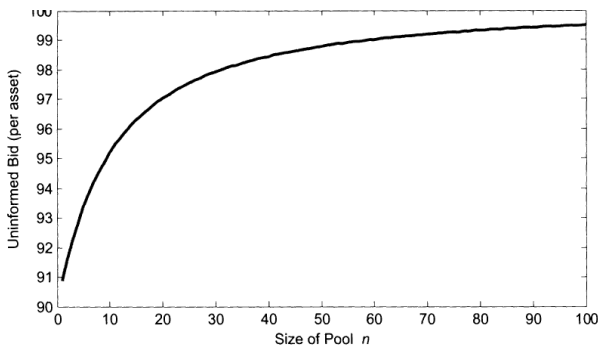
(b) Figure 2: payoff by pool size and correlation.

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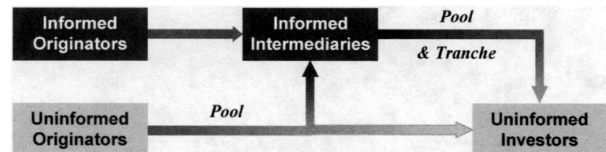
- Figure 1 plots expected issuer payoff per asset against the face value of debt per asset.
- For $n = 1$, the payoff is highest at a very large face value, equivalent to selling a pass-through security.
- For intermediate pool sizes, pooling can reduce payoff because information destruction is strong.
- For large pools, the optimal debt face value moves toward the worst-case value, and payoff rises as residual risk diversifies.
- Figure 2 shows that pooling helps when private information is common across assets and residual risk is diversifiable.
- If residual risk is almost perfectly correlated, pooling hurts because diversification is absent while information destruction remains.

Economic message. Pooling is not valuable by itself. It becomes valuable when tranching converts diversified residual risk into a senior, information-insensitive claim.

5.2 Figure 3 + Figure 4: Uninformed pooling and intermediation flow



(a) Figure 3: uninformed bid by pool size.



(b) Figure 4: asset flow over time.

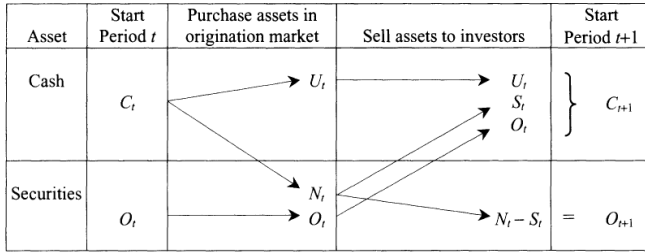
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- Figure 3 shows that the uninformed bid per asset increases rapidly with pool size.
- The bid starts well below expected value because informed buyers cherry-pick good assets.

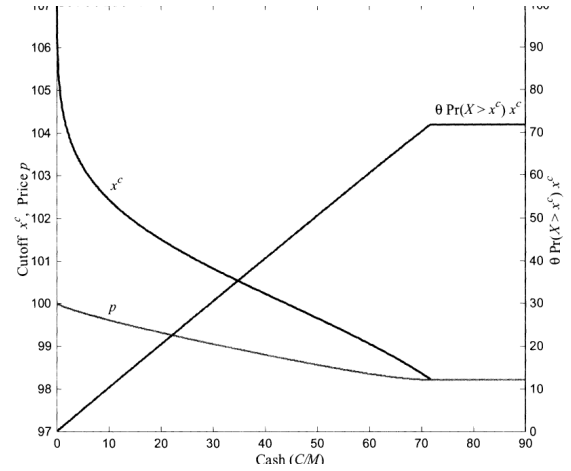
- As the pool grows, cherry-picking becomes less precise, and the bid approaches expected value.
- Figure 4 summarizes the dynamic intermediation chain: originators create pools, informed intermediaries acquire assets or pools, and intermediaries pool and tranche for uninformed investors.

Economic message. The paper’s two pooling results are complementary. Uninformed originators pool to reduce adverse selection; informed intermediaries later pool and tranche to create liquid senior claims.

5.3 Figure 5 + Figure 6: Dynamic timing and cash-constrained acquisition



(a) Figure 5: timing of the dynamic model.



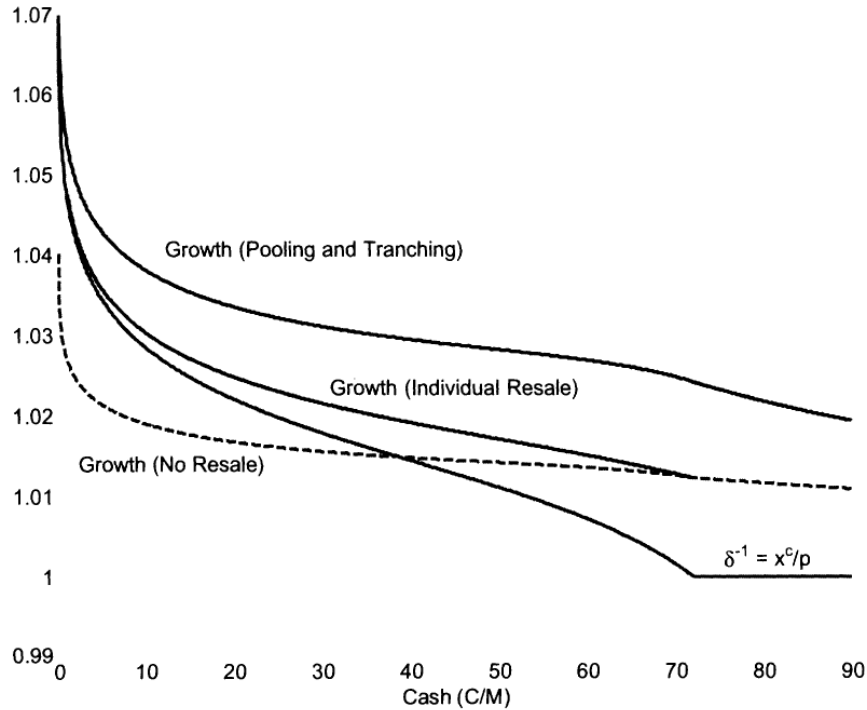
(b) Figure 6: cutoff, price, and worst-case portfolio value.

Read:

- Figure 5 shows the intermediary’s period-by-period balance sheet transformation.
- Cash C_t is used to buy new assets N_t in the origination market.
- Sales to investors generate cash S_t , and retained assets become next period’s old securities.
- Figure 6 shows how intermediary cash relative to market size affects the acquisition equilibrium.
- As intermediary cash grows, the informed cutoff x^c and price p fall until the cash constraint stops binding.
- The worst-case value of the purchased portfolio rises with intermediary size and then plateaus.

Economic message. Capital availability changes adverse selection in the origination market. More intermediary capital allows more informed buying, but resale through securitization is needed to keep capital turning over.

5.4 Figure 7: Intermediary growth and resale technology



Original paper Figure 7: intermediary growth rate under pooling and tranching, individual resale, and no resale.

Read:

- Pooling and tranching gives the highest intermediary growth rate across a wide range of intermediary sizes.
- Individual resale improves on no resale but does not scale as effectively.
- The no-resale strategy limits the intermediary's ability to recycle capital.
- The graph illustrates Theorem 7: securitization allows informed intermediaries to grow faster and hold a larger market share.

Economic message. Pooling and tranching leverages information. It lets intermediaries repeatedly monetize low-information-sensitive claims and redeploy cash into newly underpriced assets.

6 Short summary

- The paper studies why assets are pooled and tranced in securitization markets.
- The key friction is asymmetric information about asset values.
- For an informed issuer, pure pooling is worse than separate sales because it destroys asset-specific information.
- The seller's payoff is convex in private asset quality because high-quality sellers optimally sell less.
- Pooling averages high- and low-quality assets and removes this option.
- Tranching changes the result by allowing the issuer to sell a low-risk senior claim and retain the information-sensitive residual.
- If residual risks diversify, large pools can support highly liquid debt tranches.

- Pooling and tranching is most valuable when private information is common across assets but residual risk is idiosyncratic.
- Multiple tranches and ex post design preserve the same intuition: the issuer sells the safest cash flows first.
- For uninformed originators, pure pooling is beneficial because it reduces informed buyers' ability to cherry-pick.
- The dynamic model combines these results: originators pool, informed intermediaries buy the best pools, and intermediaries further pool and tranche to resell assets.
- Securitization allows intermediaries to recycle capital and earn higher returns to private information.
- The paper provides a theoretical explanation for pass-through MBS pools, REMICs, CMOs, CBOs, and related structured products.

7 Conclusion

7.1 Core conclusion

DeMarzo's central conclusion is that pooling and tranching can be optimal under asymmetric information, but pooling alone is not automatically beneficial. Informed sellers dislike pure pooling because it destroys the value of asset-specific information. They benefit from pooling only when tranching allows them to sell a senior claim whose value is protected by diversification.

7.2 Economic intuition

The intuition is a trade-off. Pooling destroys information because it turns distinct asset qualities into one average pool quality. But pooling also diversifies residual risk. Tranching lets the issuer exploit the second force without suffering too much from the first: the issuer sells the safe, liquid, senior part and retains the information-sensitive residual. Uninformed originators have the opposite pooling motive: pooling prevents informed buyers from selecting only the best assets.

7.3 Takeaway

The paper turns securitization into a theory of informed intermediation. Originators pool because they are uninformed; intermediaries buy because they are informed; intermediaries tranche because they need liquidity without giving up too much informational rent. The enduring lesson is that structured finance can arise endogenously as a response to information problems, not merely as regulatory arbitrage or transaction-cost minimization.